

SOFTWARE REQUIREMENTS SPECIFICATION

SRM Sports Council Website

Field	Details
Document Title	Software Requirements Specification
Project	SRM Sports Council Website
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the SRM Sports Council website. It is intended to serve as the definitive reference for the development team, project stakeholders, and quality assurance personnel throughout the software development lifecycle.

1.2 Project Overview

The SRM Sports Council requires a professional, public-facing website to serve as a central hub for all sports-related activities, club information, event management, and achievement showcases within the university. The platform will replace fragmented communication channels and provide a single authoritative source of information for students, faculty, and administrators.

1.3 Scope

The website will consist of four primary pages with supporting sub-sections:

- Home Page — High-impact landing page with dynamic statistics, news carousel, and council hierarchy
- Clubs Page — Sports club directory with individual club detail views
- Events Page — Chronological event management with live, upcoming, and archived events
- Achievements Page — Trophy cabinet showcasing trophies, accolades, and individual records

1.4 Definitions and Acronyms

Term / Acronym	Definition
SRS	Software Requirements Specification
SRM / SRMAP	SRM University AP — the client institution
UI	User Interface
CMS	Content Management System
API	Application Programming Interface
CDN	Content Delivery Network
SSL/TLS	Secure Sockets Layer / Transport Layer Security
WCAG	Web Content Accessibility Guidelines

1.5 References

- SRM Sports Council — Website Initial Plan (source document)
- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications
- WCAG 2.1 Accessibility Guidelines

2. Overall Description

2.1 Product Perspective

The SRM Sports Council website is a standalone web application hosted on a publicly accessible web server. It will integrate with a backend content management layer to allow authorised administrators to update content such as news, event results, club rosters, and achievements without requiring developer intervention.

2.2 Product Functions — Summary

#	Function Area	Summary
F-01	Home Page	Hero section, dynamic stats, news image carousel, council hierarchy
F-02	Clubs Directory	Sport tile grid with individual club popup/detail pages
F-03	Events Management	Live, upcoming, and archived events with result cards
F-04	Achievements Showcase	Trophy gallery, inter-university accolades, individual athlete records
F-05	Admin / CMS	Backend panel for authorised content updates
F-06	Hosting & Deployment	Reliable, secure, scalable public hosting infrastructure

2.3 User Classes

- General Visitors — Students, parents, and public browsing sports information (read-only)
- Registered Students — Students who register/sign up for sports events
- Club Representatives — Convenors and co-convenors who manage club content
- Council Administrators — Directors who manage the entire website content
- System Administrators — IT staff responsible for hosting and deployment

2.4 Operating Environment

- The website must be accessible from any modern web browser (Chrome, Firefox, Safari, Edge)
- The website must be fully responsive across desktop, tablet, and mobile screen sizes
- The website will be hosted on a cloud or dedicated web server with a domain name
- HTTPS/SSL certificate must be enabled at all times

2.5 Design and Implementation Constraints

- The system must use widely supported web technologies (HTML5, CSS3, JavaScript)
- Images and media must be optimised for fast loading on mobile networks
- All dynamic data (stats, team lists, results) must be manageable via a CMS or admin panel
- The website must comply with WCAG 2.1 Level AA accessibility standards

3. Functional Requirements

3.1 Page 1: Home Page

The Home Page is the primary entry point and must immediately communicate the scale and prestige of the SRM Sports Council.

3.1.1 Hero Section

Req ID	Requirement	Priority
FR-HP-01	Display a large, prominent main title 'SRM Sports Council' using bold, high-impact typography	HIGH
FR-HP-02	Display a dynamic Stats Bar immediately below the title showing Total Teams count and Total Members count, populated from the backend	HIGH
FR-HP-03	Stats counters must animate on page load (count-up animation)	MEDIUM
FR-HP-04	The hero section must feature a full-width background (image or video) reflecting sports activity	HIGH

3.1.2 News Ticker / Image Carousel

Req ID	Requirement	Priority
FR-HP-05	Display an image carousel showing high-quality action photographs from recent games	HIGH
FR-HP-06	Each carousel slide must support an overlaid 'Latest News' headline text	HIGH
FR-HP-07	The carousel must support auto-play with manual navigation controls (prev/next and dot indicators)	MEDIUM
FR-HP-08	Carousel images must be manageable via the admin panel (add, remove, reorder)	HIGH

3.1.3 About Council & Core Values Section

3.2 Page 2: Clubs

The Clubs page serves as a complete directory of all sporting disciplines under the SRM Sports Council.

Req ID	Requirement	Priority
FR-HP-09	Display a dedicated 'About the Council' section with the council's mission and core values	HIGH
FR-HP-10	Display the Council Hierarchy with names, titles, and photographs for Directors, Convenors, Coaches, and the Student Body	HIGH
FR-HP-11	The hierarchy section must be structured in clearly labelled tiers	MEDIUM
FR-HP-12	Council member details (name, title, photo) must be editable via the admin panel	HIGH

3.2.1 Club Grid Display

Req ID	Requirement	Priority
FR-CL-01	Display all sports clubs as clickable cards/tiles in a responsive grid layout	HIGH
FR-CL-02	Initial clubs to include: Cricket, Badminton, Tennis, Gym, Carroms, Chess, and at least two expansion slots	HIGH
FR-CL-03	Each club tile must display the sport's icon/logo and name	HIGH
FR-CL-04	The grid must reflow responsively across all screen sizes	HIGH
FR-CL-05	Administrators must be able to add new clubs to the grid via the admin panel	MEDIUM

3.2.2 Individual Club Detail View (Popup or Dedicated Page)

Req ID	Requirement	Priority
FR-CL-06	Clicking a club tile must open a detailed club view (modal popup or dedicated sub-page)	HIGH
FR-CL-07	The club detail view must display the Convenor and Co-convenor names and photographs	HIGH
FR-CL-08	The club detail view must display the current Team Roster as a list or table	HIGH
FR-CL-09	The club detail view must include a gallery section showing sport-specific photographs	HIGH
FR-CL-10	Club content (roster, leadership, gallery) must be editable by authorised users via the admin panel	HIGH

3.3 Page 3: Events

The Events page provides a chronological view of all sports events, segmented into live, upcoming, and past categories.

3.3.1 Sub-Section A: Present / Live Events

Req ID	Requirement	Priority
FR-EV-01	Display a prominently highlighted 'Live Now' section showing currently active events	HIGH
FR-EV-02	Show live scores or ongoing tournament banners for active fixtures	HIGH
FR-EV-03	Live section must auto-refresh or update in near real-time (polling or websocket)	MEDIUM

3.3.2 Sub-Section B: Future / Upcoming Events

Req ID	Requirement	Priority
FR-EV-04	Display a calendar view or list view of all upcoming fixtures	HIGH
FR-EV-05	Each upcoming event entry must show event name, sport, date, time, and venue	HIGH
FR-EV-06	Each upcoming event must include a registration link where applicable	MEDIUM
FR-EV-07	Events must be sortable and filterable by sport type and date	MEDIUM

3.3.3 Sub-Section C: Past Events / Archive

Req ID	Requirement	Priority
FR-EV-08	Maintain a searchable archive database of all previous tournaments and events	HIGH
FR-EV-09	Each past event must be displayed as a Result Card containing: Event name, Game details, Gold/1st place winner, Silver/2nd place runner-up, and additional placement details	HIGH
FR-EV-10	Result Cards must support photos from the event	MEDIUM
FR-EV-11	Archive must support pagination or infinite scroll for large numbers of historical events	MEDIUM

3.4 Page 4: Achievements

The Achievements page functions as the trophy cabinet and hall of fame for the SRM Sports Council.

Req ID	Requirement	Priority
FR-AC-01	Display a visually engaging gallery of trophies won by SRMAP teams using a Masonry Wall or Timeline layout	HIGH

Req ID	Requirement	Priority
FR-AC-02	Include a section for inter-university accolades with event names, years, and outcomes	HIGH
FR-AC-03	Include an Individual Student-Athlete Records section highlighting outstanding personal achievements	HIGH
FR-AC-04	Each achievement entry must support a photo, title, description, date, and sport category	HIGH
FR-AC-05	Achievements content must be fully manageable via the admin panel	HIGH
FR-AC-06	Display achievements in reverse-chronological order by default with filter by sport and year	MEDIUM

4. Admin Panel & Content Management

4.1 Authentication

- FR-AD-01: The admin panel must be secured with username/password authentication
- FR-AD-02: Support role-based access — Super Admin (full access), Club Rep (own club only)
- FR-AD-03: Session timeout must be enforced after a period of inactivity

4.2 Content Management Capabilities

Content Area	Admin Actions Required
Hero Stats	Edit Total Teams and Total Members counts
News Carousel	Add / remove / reorder slides and headline text
Council Hierarchy	Add / edit / remove member names, titles, and photos
Clubs	Add new clubs, edit club details, manage rosters and galleries
Events	Create / edit / delete events across all three categories; post live scores
Achievements	Add / edit / delete achievement entries with photos

5. Non-Functional Requirements

5.1 Performance

- NFR-PE-01: Page load time must not exceed 3 seconds on a standard broadband connection
- NFR-PE-02: All images must be compressed and served via a CDN where applicable
- NFR-PE-03: The website must support a minimum of 500 simultaneous users without performance degradation

5.2 Security

- NFR-SE-01: All traffic must be encrypted using HTTPS/TLS (SSL certificate mandatory)
- NFR-SE-02: The admin panel must be protected against common web vulnerabilities (SQL injection, XSS, CSRF)
- NFR-SE-03: User-uploaded media must be scanned and validated before storage
- NFR-SE-04: Database credentials and API keys must not be exposed in client-side code

5.3 Availability & Reliability

- NFR-AV-01: The website must achieve a minimum uptime of 99.5% per month
- NFR-AV-02: Automated backups of all content and database data must run daily
- NFR-AV-03: A restore procedure must be documented and tested

5.4 Usability & Accessibility

- NFR-US-01: The website must comply with WCAG 2.1 Level AA accessibility standards
- NFR-US-02: Navigation must be intuitive and require no more than three clicks to reach any content
- NFR-US-03: The website must be fully responsive on mobile (320px+), tablet (768px+), and desktop (1280px+) viewports

5.5 Maintainability

- NFR-MA-01: All code must be well-commented and follow a consistent coding standard
- NFR-MA-02: The system must support content updates without requiring a code deployment
- NFR-MA-03: Deployment must be documented with clear setup instructions for a new developer

6. Hosting Requirements

6.1 Infrastructure Specifications

Component	Requirement
Hosting Type	Cloud VPS or managed hosting (e.g., AWS, DigitalOcean, Hostinger, or equivalent)
Web Server	Apache or Nginx
Operating System	Linux-based (Ubuntu 22.04 LTS or equivalent)
Database	MySQL 8.0+ or PostgreSQL 14+ for structured content
Storage	Minimum 20 GB SSD storage; scalable media storage (S3-compatible recommended)
Bandwidth	Minimum 1 TB/month unmetered or sufficient for expected traffic
SSL Certificate	Let's Encrypt (free) or commercial SSL — auto-renewing
Domain	Custom domain (e.g., sportscouncil.srmuniversity.ac.in or equivalent)
Backup	Daily automated backups with minimum 14-day retention

6.2 Deployment Requirements

- A staging/test environment must be available separate from production
- A CI/CD pipeline or documented manual deployment process must be in place
- DNS records must be correctly configured for the domain, www subdomain, and any required mail/verification records
- A monitoring and alerting tool must be configured to notify administrators of downtime or errors

6.3 Recommended Technology Stack

Layer	Technology Options
Frontend	React.js / Next.js OR plain HTML5 + CSS3 + JavaScript
Backend / CMS	Node.js + Express OR WordPress with custom theme OR Strapi headless CMS
Database	MySQL / PostgreSQL / MongoDB
Media Storage	AWS S3 / Cloudflare R2 / Local server storage

Layer	Technology Options
Hosting	AWS EC2 / DigitalOcean Droplet / Hostinger VPS / cPanel shared hosting
CDN	Cloudflare (recommended for performance and DDoS protection)
SSL	Let's Encrypt via Certbot (auto-renewing, free)

7. System Constraints & Assumptions

7.1 Constraints

1. The website is intended for SRM University AP's Sports Council only and will not handle financial transactions.
2. Live score updates are limited to tournaments actively managed by the council; third-party live sports data feeds are out of scope.
3. The admin panel is restricted to council-authorized personnel; no public registration is required for the admin module.
4. Media file uploads are limited to image formats (JPEG, PNG, WebP, GIF) and standard video formats (MP4) if video support is included.

7.2 Assumptions

5. The SRM Sports Council will provide all initial content (member details, club information, photos, etc.) prior to launch.
6. A designated system/web administrator from the council or university IT department will manage ongoing hosting.
7. The council will nominate at least one Super Admin and club-level representatives to maintain content after handover.
8. Domain registration and hosting costs are to be borne by SRM University AP or the Sports Council.

8. Requirements Traceability Matrix

Req ID	Requirement Description	Page / Module	Priority
FR-HP-01 – 04	Hero section with title and dynamic stats	Home	HIGH
FR-HP-05 – 08	News image carousel with headlines	Home	HIGH
FR-HP-09 – 12	About section and council hierarchy	Home	HIGH
FR-CL-01 – 05	Clubs grid display	Clubs	HIGH
FR-CL-06 – 10	Individual club detail view	Clubs	HIGH
FR-EV-01 – 03	Live / present events section	Events	HIGH
FR-EV-04 – 07	Upcoming events with calendar/list	Events	HIGH
FR-EV-08 – 11	Past events archive with result cards	Events	HIGH
FR-AC-01 – 06	Achievements and trophy showcase	Achievements	HIGH
FR-AD-01 – 03	Admin authentication and roles	Admin Panel	HIGH
NFR-PE-01 – 03	Performance requirements	Infrastructure	HIGH
NFR-SE-01 – 04	Security requirements	Infrastructure	HIGH
NFR-AV-01 – 03	Availability and backup requirements	Hosting	HIGH
NFR-US-01 – 03	Usability and accessibility	All Pages	MEDIUM
NFR-MA-01 – 03	Maintainability	All Modules	MEDIUM

9. Appendix — Page Structure Summary

Page	Key Sections	Key Features
Home	Hero, Stats Bar, Carousel, About & Hierarchy	Dynamic stats, animated counters, news slider
Clubs	Sport Grid, Club Detail View	Clickable tiles, roster, gallery, leadership profiles
Events	Live Events, Upcoming, Archive	Real-time scores, calendar view, result cards
Achievements	Trophy Gallery, Accolades, Athlete Records	Masonry/timeline layout, filter by sport & year
Admin Panel	Login, Dashboard, Content Editors	Role-based access, media upload, content CRUD

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